What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Development of the Gastrointestinal System

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Adapted from: Development of the Gastrointestinal System

Listen to this Brick
0:00 / 25:14 1x

Learning Objectives (3) Versions

After completing this brick, you will be able to:

1. Outline the early formation of the GI tract tube.
2. Describe the development of the foregut, midgut, and hindgut.
3. Outline the process of recanalization in the GI tract tube and intraperiton[illegible]bdominal organs.

CASE CONNECTION

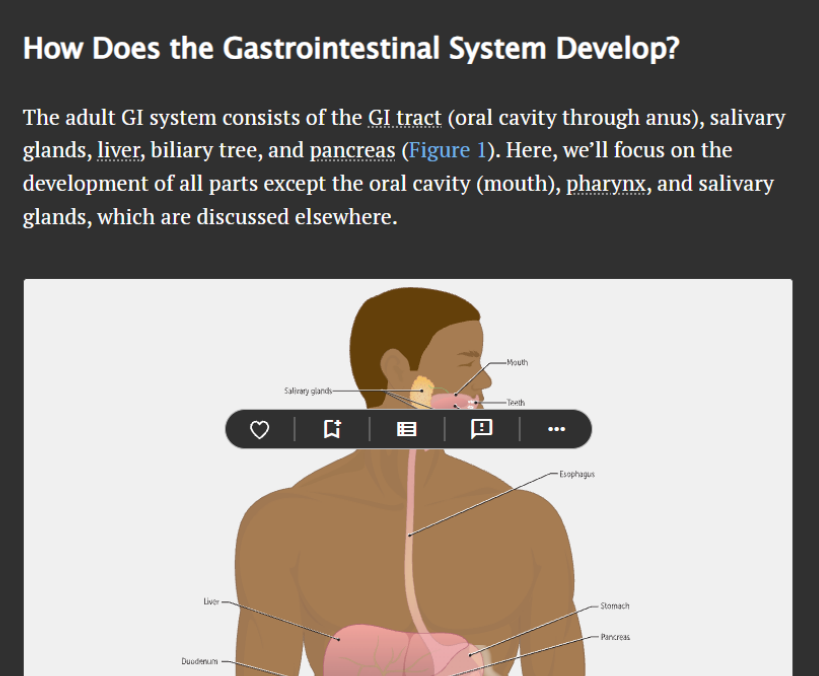
You have just delivered a 7-lb 5-oz newborn with APGAR scores of 8/10. The mother had an uncomplicated pregnancy and delivery. The patient does well for the first day but then develops abdominal distension and several episodes of bilious vomiting. "Was it something that we did?" asks their parent.

How will you explain the diagnosis and treatment to the patients' parents? Consider your answer as you read, and we'll revisit it at the end of the brick.

GO TO CONCLUSION ↓

How Does the Gastrointestinal System Develop?

The adult GI system consists of the GI tract (oral cavity through anus), salivary glands, liver, biliary tree, and pancreas (Figure 1). Here, we'll focus on the development of all parts except the oral cavity (mouth), pharynx, and salivary glands, which are discussed elsewhere.


Salivary glands
Mouth
Teeth
Esophagus
Liver
Stomach
Pancreas
Duodenum
Gallbladder
Transverse colon
Ascending colon
Jejunum
Descending colon
Ileum
Appendix
Sigmoid colon
Rectum
Anus

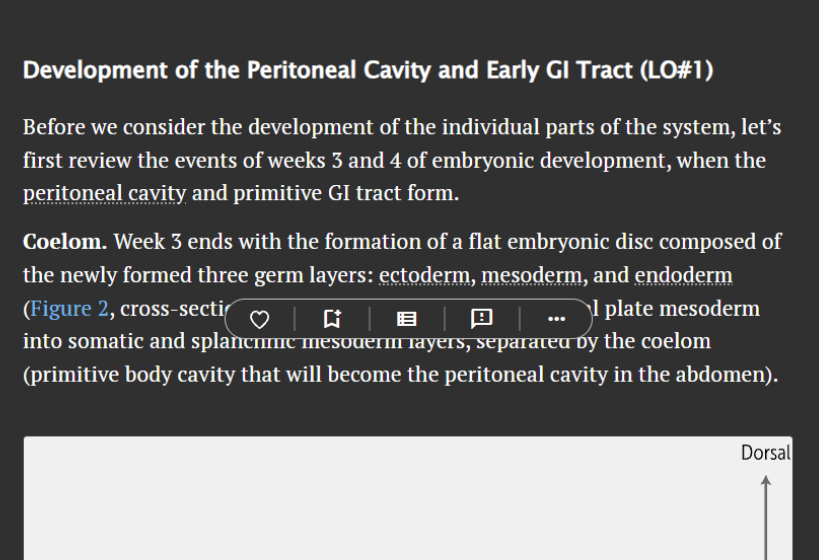
QUIZ

Figure 1

Development of the Peritoneal Cavity and Early GI Tract (LO#1)

Before we consider the development of the individual parts of the system, let's first review the events of weeks 3 and 4 of embryonic development, when the peritoneal cavity and primitive GI tract form.

Coelom. Week 3 ends with the formation of a flat embryonic disc composed of the newly formed three germ layers: ectoderm, mesoderm, and endoderm (Figure 2, cross-secti[illegible]l plate mesoderm into somatic and splanchnic mesoderm layers, separated by the coelom (primitive body cavity that will become the peritoneal cavity in the abdomen).

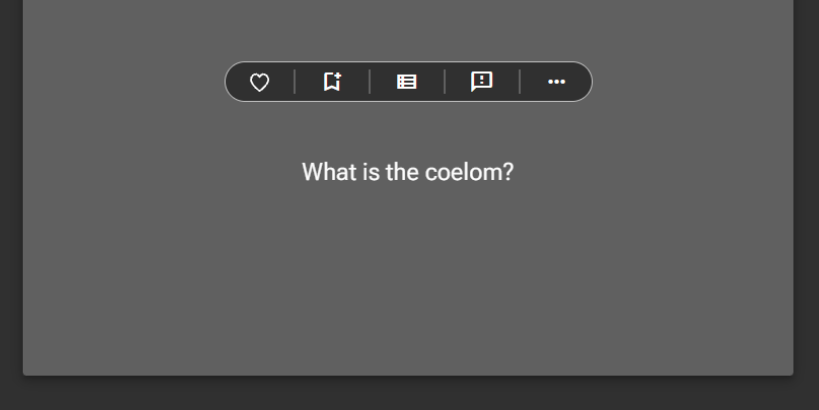

Ectoderm
Coelom
Endoderm
Intermediate mesoderm
Somite
Neural folds
Somatic mesoderm
Splanchnic mesoderm
Lateral plate
Dorsal
Ventral

QUIZ

Figure 2

What is the coelom?

Lateral Folding. Lateral folding during week 4 creates the body trunk and its contents—the primitive GI tract (a single, midline tube) surrounded by the peritoneal cavity (Figure 3, cross-section). Note that the GI tract tube is formed from endoderm (forms the lining) and splanchnic mesoderm (forms the rest of the wall).


Ectoderm
Coelom
Endoderm
Intermediate mesoderm
Somite
Neural folds
Somatic mesoderm
Splanchnic mesoderm
Lateral plate
Aorta
Embryonic coelom
Amnion
Yolk sac
GI tract
Dorsal mesentery
Splanchnic mesoderm layer
Somatic mesoderm layer
Ventral mesentery disappearing
Peritoneal cavity
Ventral body wall

QUIZ

Figure 3

Ventral and Dorsal Mesenteries. The abdominal GI tube is initially attached to the ventral and dorsal abdominal (peritoneal) walls by thin, midline mesenteries that are formed from mesoderm (see Figure 3, right). The ventral mesentery will largely disappear, whereas the dorsal mesentery will largely persist. It is needed as an attachment of the GI tract to the posterior abdominal (peritoneal) wall and also as a road through which nerves and vessels travel to the GI tract.

Foregut, Midgut, and Hindgut (LO#2)

Now let's look at the GI tract inside the embryo in a midsagittal view (Figure 4A/B). We can see that the single, long, midline GI tract subdivides into foregut, midgut, and hindgut regions.

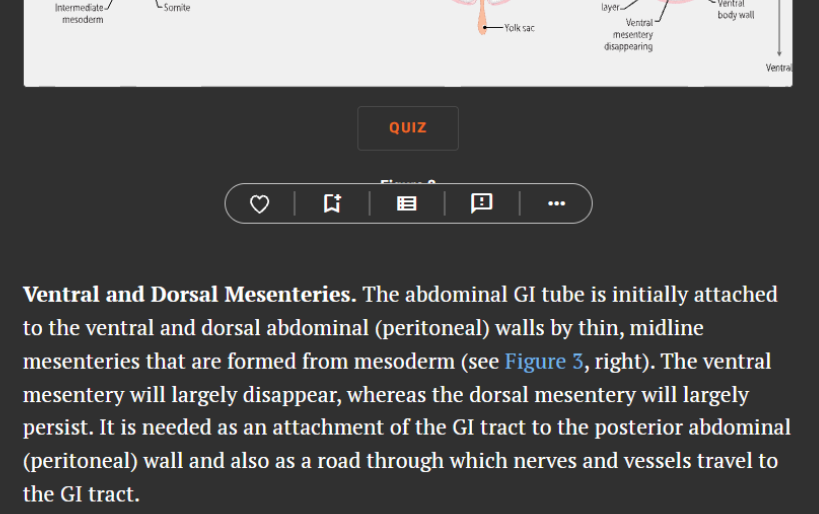

Amniotic sac
Midgut
Vitelline duct
Neural tube
Foregut
Heart
Hindgut
Cloacal membrane
Yolk sac
Allantois
Ectoderm
Mesoderm
Endoderm

Figure 4A

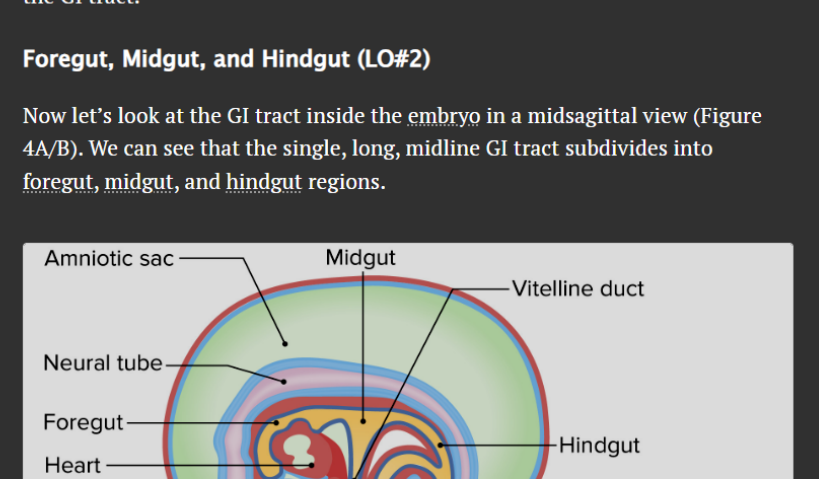

Pharyngeal origin
Foregut
Midgut
Hindgut
Celiac trunk
Aorta
Superior mesenteric artery
Inferior mesenteric artery

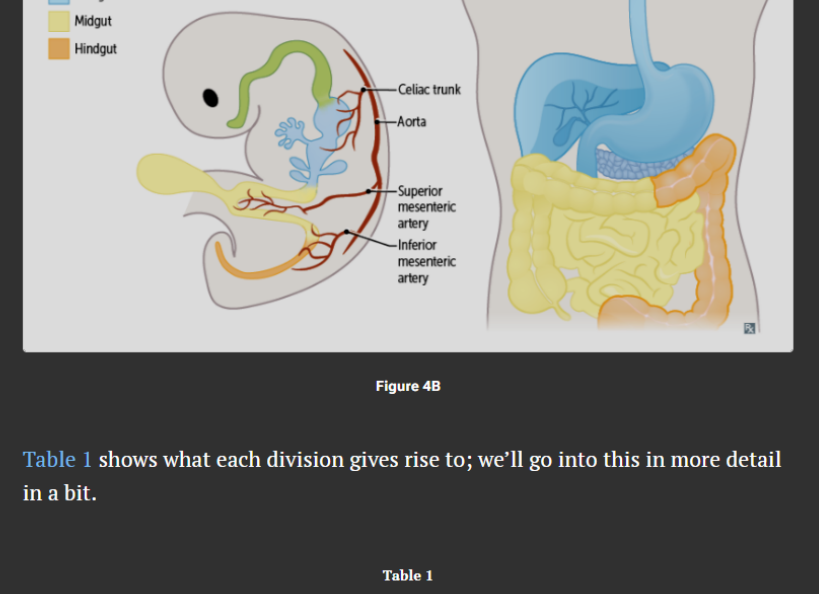
Figure 4B

Table 1 shows what each division gives rise to; we'll go into this in more detail in a bit.

Table 1

Foregut	Midgut	Hindgut
Pharynx Esophagus Stomach Proximal duodenum Pancreas Liver Gallbladder Bile ducts	[illegible] Ileum Cecum Ascending colon Transverse colon (proximal two-thirds)	[illegible]sverse colon (distal one-third) Descending colon Sigmoid colon Rectum Proximal anal canal

Blood Supply. As the more detailed Figure 5 shows, the distal foregut (and its derivatives) is supplied by the celiac artery (trunk). The midgut (and its derivatives) is supplied by the superior mesenteric artery (SMA). The hindgut (and its derivatives) is supplied by the inferior mesenteric artery (IMA).


Cranial
Caudal
Pharynx
Celiac trunk
Septum transversum
Aortic arches
Spinal cord
Superior mesenteric artery
Inferior mesenteric artery
Brain
Midgut
Heart
Yolk stalk
Cloaca
caudal part of hindgut

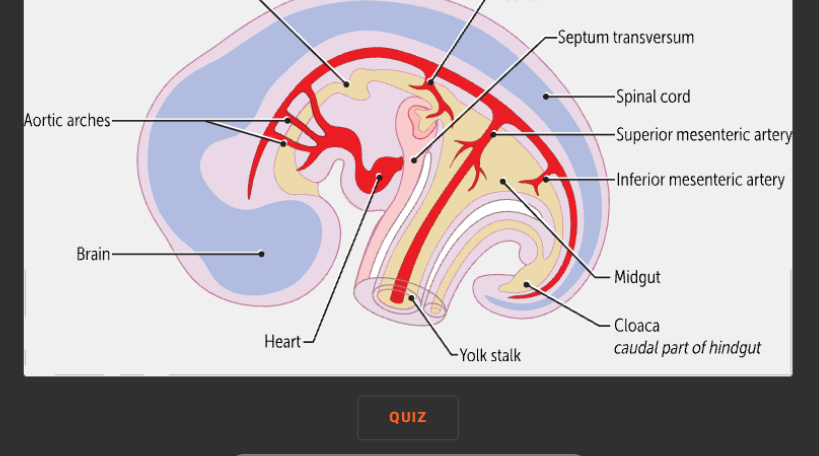
QUIZ

How Do Recanalization and Organ Positioning Drive Further GI Tract Development? (LO#3)

It's helpful at this point to consider two basic developmental processes that occur in multiple GI structures during development: recanalization and intraperitoneal and retroperitoneal positioning of abdominal organs.

Recanalization

Recanalization is an interesting stage in the development of most of the GI tract tube, although we don't know why it occurs. In most regions, early in development, the epithelium (endoderm) of the tube proliferates so that it fills and obstructs the lumen (Figure 6). This is followed by removal of the cells in the lumen to recreate an open (patent) tube.


Hollow gut tube
Gut tube occluded by endodermal proliferation
Recanalization
Definitive hollow gut

CLINICAL CORRELATION

If the cell removal step in a region is incomplete, a narrowed lumen results, called stenosis (Figure 7). When cell removal fails to occur in a region, it results in obstruction, called atresia.

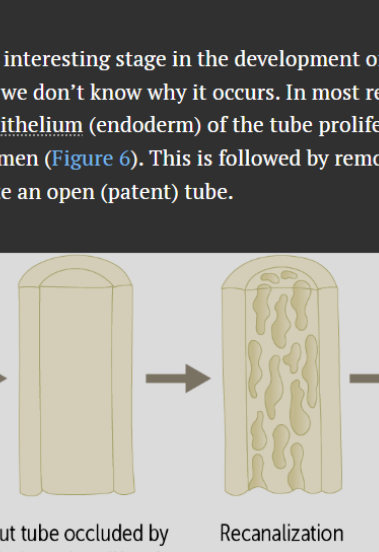

Stenosis
Atresia

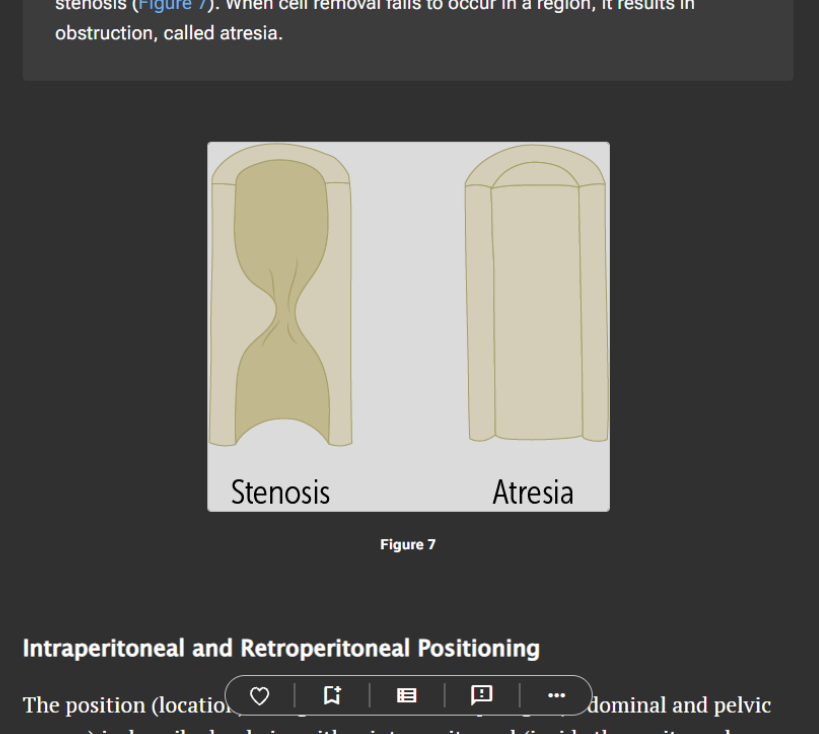
Figure 7

Intraperitoneal and Retroperitoneal Positioning

The position (locatio[illegible]dominal and pelvic organs) is described as being either intraperitoneal (inside the peritoneal cavity) or retroperitoneal (outside the peritoneal cavity). Let's consider how these positions are acquired during development.

Remember that lateral folding creates the abdominal GI tract and the surrounding peritoneal cavity. The retroperitoneal organs arise outside the peritoneal cavity and include the kidneys, adrenal glands, and the urinary bladder. Also recall that the primitive abdominal GI tract is attached to the abdominal (peritoneal) walls by the dorsal and ventral mesenteries (see Figure 3). By the end of development, all abdominal organs that are still attached to a mesentery are intraperitoneal and include much of the abdominal GI tract.

Secondarily Retroperitoneal Organs. Although the entire primitive abdominal GI tract is attached to the dorsal mesentery, in some regions, this mesentery will disappear due to fusion with the abdominal (peritoneal) wall (Figure 8). Those regions of the GI tract are considered to be "secondarily retroperitoneal" because they begin as intraperitoneal but later come to lie outside the peritoneal cavity. These organs include the ascending and descending colon, duodenum, and pancreas.


Aorta
Intermediate mesoderm
Splanchnic mesoderm
Somatic mesoderm
Gut
Intraembryonic coelom
Dorsal mesentery
Mesonephros (embryonic kidney)
Peritoneal cavity
Bladder
Retroperitoneal
Intraperitoneal
GI tract becoming retroperitoneal

QUIZ

Figure 8

What's the difference between an organ that is retroperitoneal and one that is secondarily retroperitoneal?

With all this in mind, let's take a look at each region of the GI tract individually, focusing on its unique development.

How Does the Foregut Develop? (Back to LO#2)

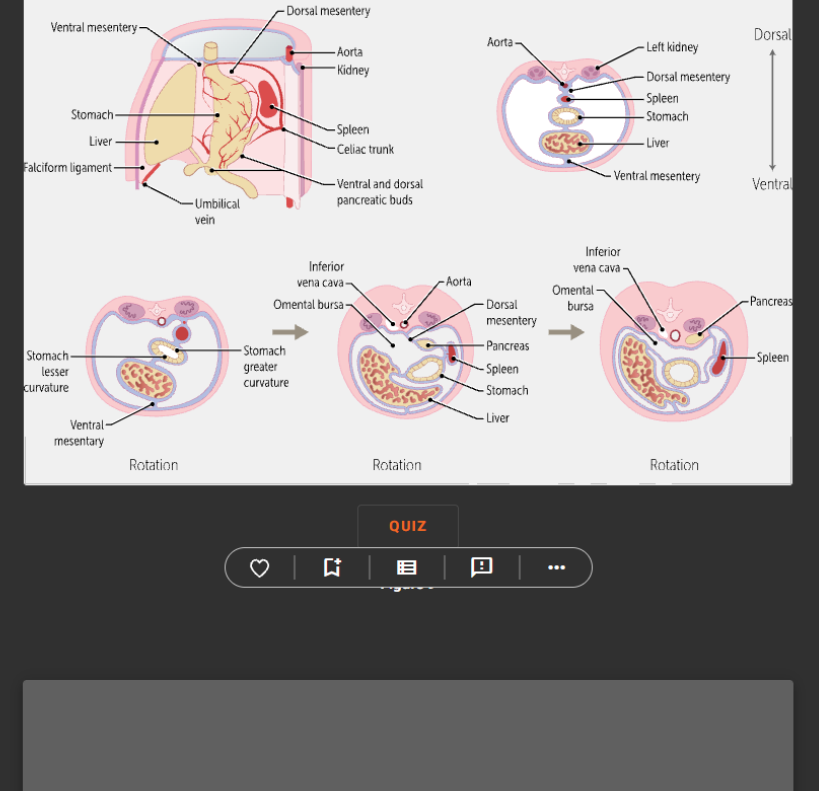
As noted, the foregut is the portion of the GI tract that spans from the mouth through the proximal duodenum, but here we're going to look at the development of just the esophagus, stomach, and proximal duodenum (and its derivatives). As each of these arises from the foregut, it is vascularized by branches of the celiac artery (trunk). Although it's not a direct foregut derivative, the spleen is also discussed here as it receives its blood supply from the celiac artery.

Esophagus

Development of the esophagus is simple. It stays in place and simply elongates. As expected, given its foregut origins, it is partially (the distal portions) vascularized by branches of the celiac trunk.

Stomach

The stomach begins development as a localized expansion of the foregut just distal to the diaphragm (Figure 9). As the stomach elongates, accelerated proliferation of the dorsal wall occurs. The faster-growing dorsal wall forms the greater curvature of the stomach, leaving the shorter ventral wall to become the lesser curvature. [illegible]dominal GI tract, the developing stomach rotates 90° clockwise in the transverse plane. This rotation accomplishes a number of important effects. First, it swings the stomach into the left upper abdominal cavity. Second, the lesser curvature ends up as the right side of the stomach, and the greater curvature ends up as the left side of the stomach. Third, a space is created posterior to the stomach called the omental bursa. It is important to remember that the GI structures form mostly anterior to posterior, and it is the rotations of the developing GI tract that take them to their final position. See Figure 9 for how that works!


Ventral
Dorsal
Dorsal mesentery
Ventral mesentery
Aorta
Kidney
Stomach
Liver
Spleen
Falciform ligament
Celiac trunk
Ventral and dorsal pancreatic buds
Umbilical vein
Left kidney
Ventral mesentery
Inferior vena cava
Omental bursa
Stomach greater curvature
Stomach lesser curvature
Pancreas
Rotation

QUIZ

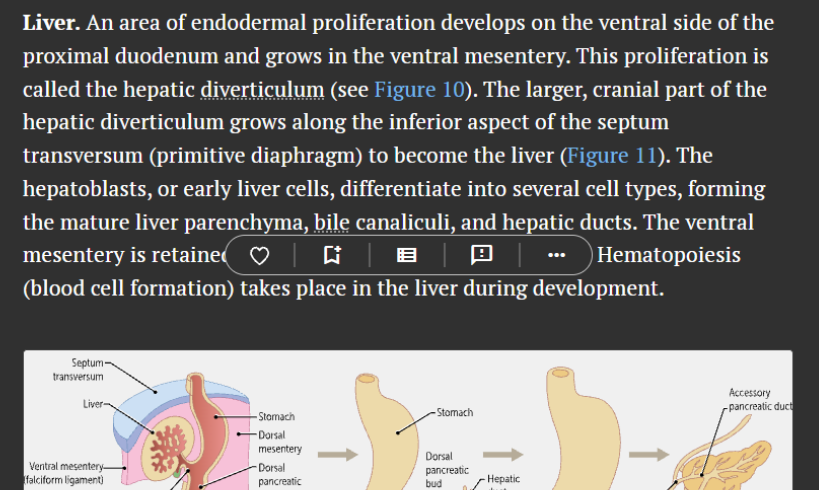
How are the greater and lesser curvatures of the stomach formed?

Proximal Duodenum

Recall that the duodenum arises from two primitive GI tract regions, the foregut and midgut. The junction of the two primitive parts of the duodenum lies just distal to the entrance of the common bile duct into the duodenum. This dual origin is important—it means that the proximal duodenum is vascularized by branches of the celiac artery, whereas the distal duodenum is vascularized by branches of the SMA.

As the original straight, midline duodenum grows long, it protrudes ventrally. Because it is attached to the stomach, the now C-shaped duodenum moves when the stomach rot[illegible]e right upper abdomen.

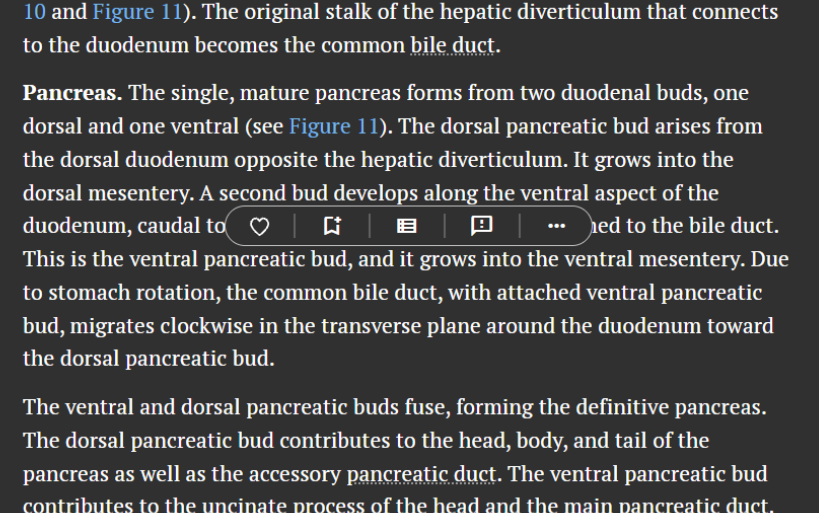
The proximal duodenum also gives rise to buds that form the liver, gallbladder, cystic duct, and pancreas. It makes sense that these would all arise from the duodenum, as they all attach to and empty their secretions into the duodenum. See Figure 10.


Peritoneal cavity
Dorsal aorta
Developing stomach
Hepatic diverticulum
Ventral mesentery
Foregut
Midgut
Duodenum
Yolk stalk
Diaphragm
Liver
Stomach
Gallbladder
Bile duct
Ventral
Dorsal

QUIZ

Figure 10

Liver. An area of endodermal proliferation develops on the ventral side of the proximal duodenum and grows in the ventral mesentery. This proliferation is called the hepatic diverticulum (see Figure 10). The larger, cranial part of the hepatic diverticulum grows along the inferior aspect of the septum transversum (primitive diaphragm) to become the liver (Figure 11). The hepatoblasts, or early liver cells, differentiate into several cell types, forming the mature liver parenchyma, bile canaliculi, and hepatic ducts. The ventral mesentery is retaine[illegible] Hematopoiesis (blood cell formation) takes place in the liver during development.


Septum transversum
Liver
Ventral mesentery (falciform ligament)
Stomach
Dorsal mesentery
Dorsal pancreatic bud
Ventral pancreatic bud
Hepatic duct
Cystic diverticulum
Gallbladder
Bile duct
Minor papilla
Major papilla
Accessory pancreatic duct
Uncinate process
Main pancreatic duct

QUIZ

Figure 11

As expected, the liver is vascularized by branches of the celiac artery (trunk).

Gallbladder and Cystic and Bile Ducts. The smaller, caudal part of the hepatic diverticulum develops into the gallbladder and cystic duct (see Figure 10 and Figure 11). The original stalk of the hepatic diverticulum that connects to the duodenum becomes the common bile duct.

Pancreas. The single, mature pancreas forms from two duodenal buds, one dorsal and one ventral (see Figure 11). The dorsal pancreatic bud arises from the dorsal duodenum opposite the hepatic diverticulum. It grows into the dorsal mesentery. A second bud develops along the ventral aspect of the duodenum, caudal to [illegible]ed to the bile duct. This is the ventral pancreatic bud, and it grows into the ventral mesentery. Due to stomach rotation, the common bile duct, with attached ventral pancreatic bud, migrates clockwise in the transverse plane around the duodenum toward the dorsal pancreatic bud.

The ventral and dorsal pancreatic buds fuse, forming the definitive pancreas. The dorsal pancreatic bud contributes to the head, body, and tail of the pancreas as well as the accessory pancreatic duct. The ventral pancreatic bud contributes to the uncinate process of the head and the main pancreatic duct. The pancreas then fuses with the dorsal abdominal (peritoneal) wall and becomes secondarily retroperitoneal. Pancreatic endodermal cells differentiate into pancreatic exocrine cells and pancreatic endocrine cells.

The pancreas is, of course, vascularized by branches of the celiac artery (trunk). What isn't explained by embryology is that the head of the pancreas is also supplied by the SMA.

Spleen

During development of the other upper abdominal organs, a cellular condensation of mesoderm develops within the midline dorsal mesentery (see Figure 9). This condensation becomes the spleen. Along with the other upper abdominal organs, the primitive spleen also rotates as the stomach rotates. This is how the spleen ends up in the left upper abdominal cavity near the stomach and pancreas.

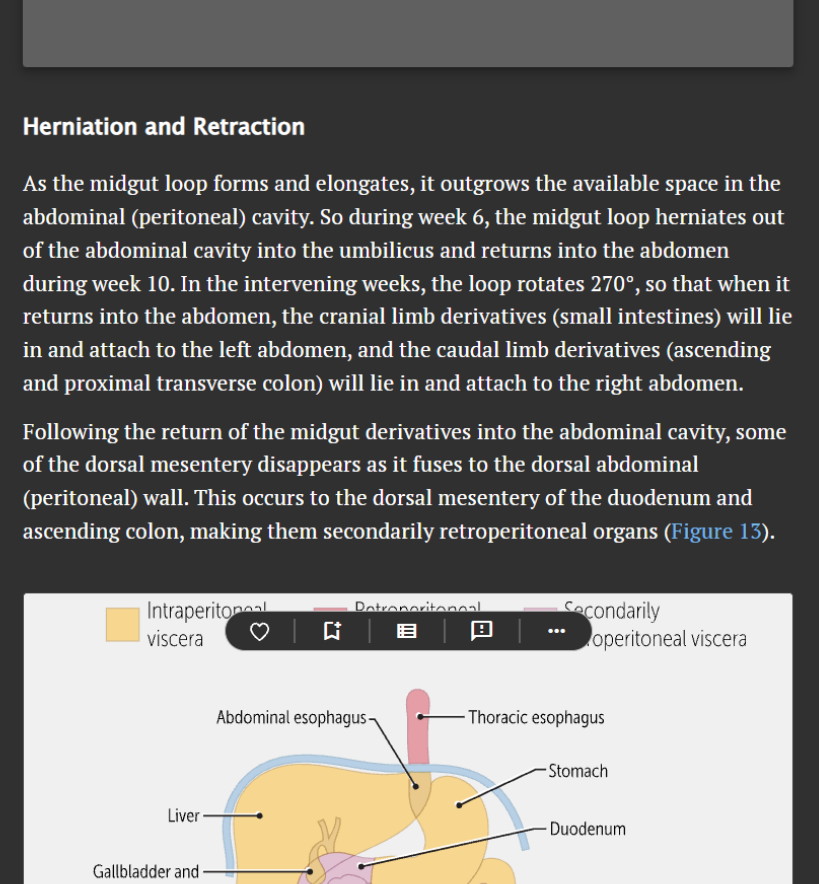
Where does the primitive spleen begin developing?

How Does the Midgut Develop?

The midgut is the middle portion of the developing GI tract. It becomes the distal duodenum, jejunum, ileum, cecum, ascending colon, and proximal two-thirds of the transverse colon. As expected, it is vascularized by branches of the SMA.

Midgut Loop

Development of the midgut is a bit complicated, but know that it mostly involves elongation and rotation, followed by placement of each derivative in the appropriate location in the abdominal (peritoneal) cavity. The primitive midgut elongates rapidly to form what is called the midgut (intestinal) loop (Figure 12). The midgut loop forms a hairpin bend at the vitelline duct (yolk stalk). The cranial limb of the loop develops numerous folds and becomes the jejunum and ileum; the caudal limb of the loop forms no folds and becomes the ascending and proximal transverse colons. Note that the SMA lies in the mesentery between t[illegible]


Ventral
Dorsal
Liver
Stomach
Gallbladder
Umbilical cord
Yolk stalk
Dorsal aorta
Spleen
Dorsal pancreatic bud
Superior mesenteric artery
Midgut loop
Inferior mesenteric artery
Ventral mesentery
Dorsal mesentery
Duodenum
Hindgut
Cecal diverticulum
Cecum
Omental bursa
Transverse colon
Lesser omentum
Descending colon
Small intestines
Sigmoid colon
Rectum
Ascending colon
Appendix and cecum

QUIZ

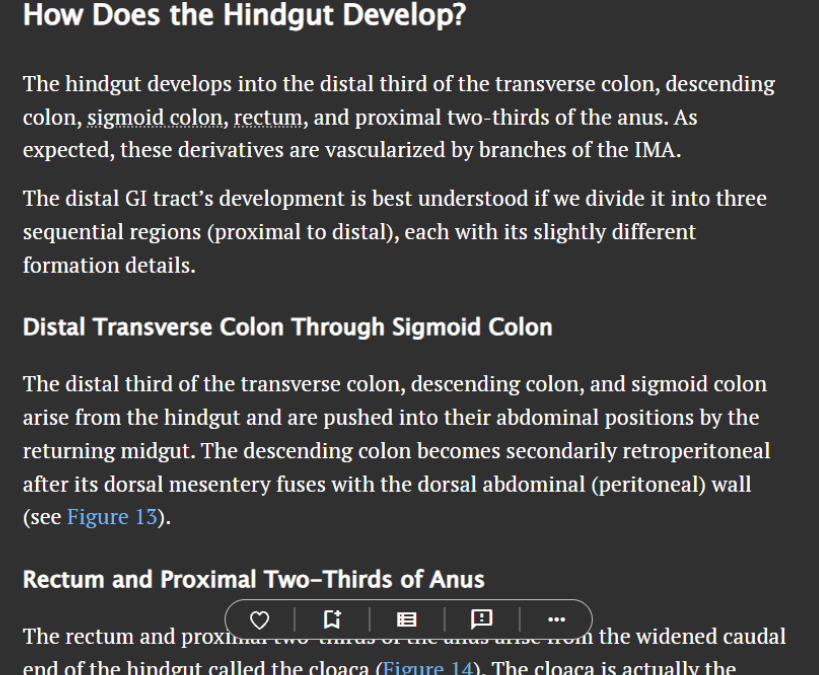
Figure 12

The cranial limb of the midgut loop develops into what GI tract regions?

Herniation and Retraction

As the midgut loop forms and elongates, it outgrows the available space in the abdominal (peritoneal) cavity. So during week 6, the midgut loop herniates out of the abdominal cavity into the umbilicus and returns into the abdomen during week 10. In the intervening weeks, the loop rotates 270°, so that when it returns into the abdomen, the cranial limb derivatives (small intestines) will lie in and attach to the left abdomen, and the caudal limb derivatives (ascending and proximal transverse colon) will lie in and attach to the right abdomen.

Following the return of the midgut derivatives into the abdominal cavity, some of the dorsal mesentery disappears as it fuses to the dorsal abdominal (peritoneal) wall. This occurs to the dorsal mesentery of the duodenum and ascending colon, making them secondarily retroperitoneal organs (Figure 13).


Intraperitoneal viscera
Retroperitoneal
Secondarily retroperitoneal viscera
Abdominal esophagus
Thoracic esophagus
Stomach
Liver
Duodenum
Gallbladder and bile duct
Pancreas
Transverse colon
Descending colon
Jejunum and ileum
Ascending colon
Cecum
Sigmoid colon
Appendix
Rectum

QUIZ

Figure 13

Why does the midgut (intestinal) loop herniate into the umbilicus?

How Does the Hindgut Develop?

The hindgut develops into the distal third of the transverse colon, descending colon, sigmoid colon, rectum, and proximal two-thirds of the anus. As expected, these derivatives are vascularized by branches of the IMA.

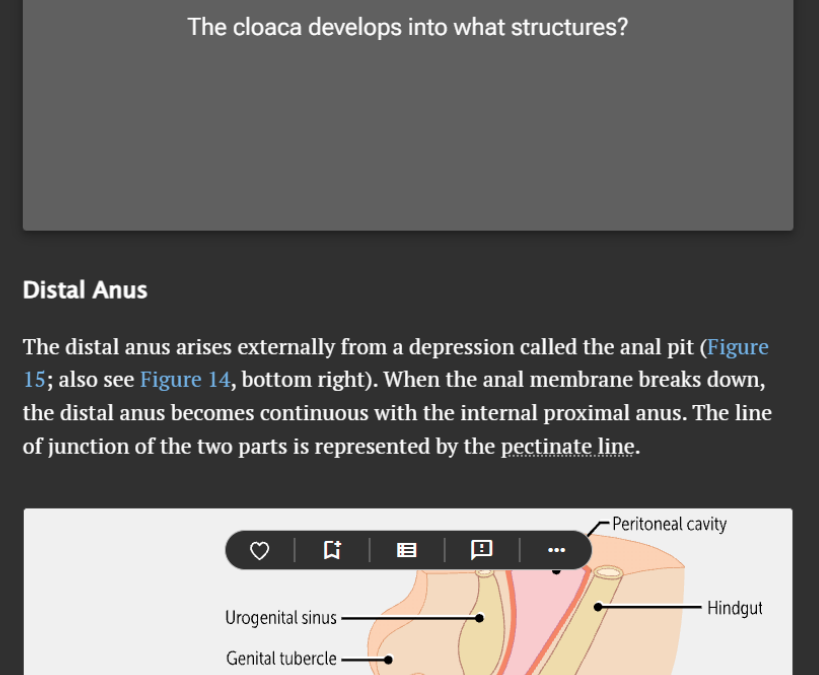
The distal GI tract's development is best understood if we divide it into three sequential regions (proximal to distal), each with its slightly different formation details.

Distal Transverse Colon Through Sigmoid Colon

The distal third of the transverse colon, descending colon, and sigmoid colon arise from the hindgut and are pushed into their abdominal positions by the returning midgut. The descending colon becomes secondarily retroperitoneal after its dorsal mesentery fuses with the dorsal abdominal (peritoneal) wall (see Figure 13).

Rectum and Proximal Two-Thirds of Anus

The rectum and proxi[illegible]m the widened caudal end of the hindgut called the cloaca (Figure 14). The cloaca is actually the common origin for the developing distal GI tract and the urinary tract. To accomplish this, the cloaca is subdivided into a ventral urinary portion and a dorsal anorectal canal by the urorectal septum. The anorectal canal becomes the rectum and proximal two-thirds of the anus, whereas the urogenital sinus becomes the bladder and pelvic urethra. The anorectal canal is separated from the exterior anal pit by the anal membrane.


Allantois
Yolk stalk (vitelline duct)
Midgut
Hindgut
Urorectal septum
Cloaca
Proctodeum
Cloacal membrane
Ventral
Dorsal
Phallus
Anal pit
Urogenital sinus
Rectum
Developing urinary bladder
Anal canal
Anal membrane
Urogenital membrane

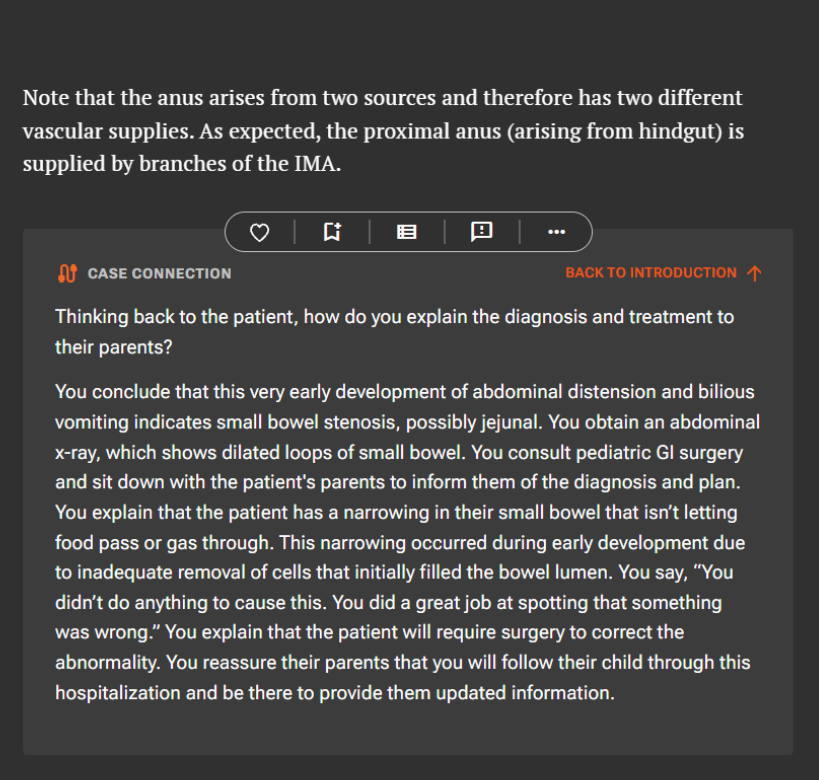
QUIZ

Figure 14

The cloaca develops into what structures?

Distal Anus

The distal anus arises externally from a depression called the anal pit (Figure 15; also see Figure 14, bottom right). When the anal membrane breaks down, the distal anus becomes continuous with the internal proximal anus. The line of junction of the two parts is represented by the pectinate line.


Peritoneal cavity
Hindgut
Urogenital sinus
Genital tubercle
Urogenital plate
Anal pit
Anal membrane
Rectum
Perineum
Pectinate line

QUIZ

Figure 15

Note that the anus arises from two sources and therefore has two different vascular supplies. As expected, the proximal anus (arising from hindgut) is supplied by branches of the IMA.

CASE CONNECTION BACK TO INTRODUCTION ↑

Thinking back to the patient, how do you explain the diagnosis and treatment to their parents?

You conclude that this very early development of abdominal distension and bilious vomiting indicates small bowel stenosis, possibly jejunal. You obtain an abdominal x-ray, which shows dilated loops of small bowel. You consult pediatric GI surgery and sit down with the patient's parents to inform them of the diagnosis and plan. You explain that the patient has a narrowing in their small bowel that isn't letting food pass or gas through. This narrowing occurred during early development due to inadequate removal of cells that initially filled the bowel lumen. You say, "You didn't do anything to cause this. You did a great job at spotting that something was wrong." You explain that the patient will require surgery to correct the abnormality. You reassure their parents that you will follow their child through this hospitalization and be there to provide them updated information.

Summary

- The gastrointestinal (GI) system develops from the embryonic coelom, which forms the peritoneal cavity, and lateral folding creates the primitive GI tract.
- The GI tract is divided into foregut, midgut, and hindgut, each giving rise to specific organs and having distinct blood supplies: celiac artery for the foregut, superior mesenteric artery for the midgut, and inferior mesenteric artery for the hindgut.
- Recanalization is a key developmental process where the GI tract tube's lumen is initially[illegible]nt tube; failure in this process can lead to stenosis or atresia.
- Intraperitoneal organs remain within the peritoneal cavity, while secondarily retroperitoneal organs initially develop intraperitoneally but later reside outside the peritoneal cavity due to mesentery fusion.
- The foregut develops into the esophagus, stomach, and proximal duodenum, with the liver, gallbladder, and pancreas arising from the proximal duodenum.
- The midgut and hindgut undergo elongation and rotation, with the midgut herniating and retracting during development, eventually forming the small intestine, colon, rectum, and anus.

Review Questions

1. **Based on embryonic development, which of the following structures is vascularized by branches of both the celiac and superior mesenteric arteries?**

A. ○ Anus
B. ● Duodenum
C. ○ Esophagus
D. ○ Ileum
E. ○ Stomach

Hide Explanation

The correct answer is duodenum (B) because it arises from both the foregut and midgut. The anus (A) arises from the hindgut and external anal pit, so it is partially supplied by the inferior mesenteric artery. The esophagus (C) arises from the foregut and is supplied by the celiac artery (trunk). The ileum (D) arises from the midgut and therefore is supplied by the superior mesenteric artery. The stomach (E) also arises from the foregut and is supplied by celiac artery (trunk).

2. **Which of the following structures is derived from the midgut?**

A. ● Distal duodenum
B. ○ Gallbladder
C. ○ Liver
D. ○ Proximal duodenum
E. ○ Stomach

Hide Explanation

The correct answer is distal duodenum (A). The distal duodenum is a derivative of the midgut. The gallbladder (B), liver (C), proximal duodenum (D), and stomach (E) are foregut derivatives. In fact, the gallbladder, liver, and pancreas are outpouchings from the proximal duodenum itself.

3. **Which of the following structures is intraperitoneal?**

A. ○ Ascending colon
B. ○ Descending colon
C. ○ Duodenum
D. ○ Pancreas
E. ● Stomach

Hide Explanation

The correct answer is stomach (E). The ascending colon (A), descending colon (B), duodenum (C), and pancreas (D) become secondarily retroperitoneal. These structures first develop within the mesentery. After the folding and rotating processes, their mesenteries fuse with the dorsal abdominal (peritoneal) wall.

4. **A 10-week-old fetus has an abnormally developed cloaca. Which of the following structures is most likely to be affected because of this anomaly?**

A. ○ Distal anus
B. ○ Liver
C. ○ Proximal transverse colon
D. ● Rectum
E. ○ Stomach

Hide Explanation

The correct answer is rectum (D). The rectum and proximal anus are the only GI tract structures that arise from the cloaca (widened part of the hindgut). The distal anus (A) arises from the external anal pit. The liver (B) arises from the foregut. The proximal transverse colon (C) arises from the midgut. The stomach (E) also arises from the foregut.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Outline~~

structures that arise from the cloaca (widened part of the hindgut). The distal anus (A) arises from the external anal pit. The liver (B) arises from the foregut. The proximal transverse colon (C) arises from the midgut. The stomach (E) also arises from the foregut.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Outline the early formation of the GI tract tube.~~
02 ☑ ~~Describe the development of the foregut, midgut, and hindgut.~~
03 ☑ ~~Outline the process of recanalization in the GI tract tube and intraperitoneal and retroperitoneal positioning of abdominal organs.~~

Objective Confidence: 3/3